SIMATS ENGINEERING
CO2-ASSSESSMENT TOOL-6
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Course Name: **CLOUD COMPUTING AND BIG DATA ANALYTICS**

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Question 1

A media company wants to migrate its video streaming platform to the cloud. The system must handle millions of concurrent users with minimal buffering. Design a cloud architecture using scalability and caching techniques. Explain how load balancing and auto-scaling can ensure smooth streaming. Discuss cost optimization strategies for such high-demand workloads.

Answer

Introduction

A video streaming platform requires high performance and continuous availability to support millions of users simultaneously. Cloud computing provides scalable infrastructure that automatically adjusts resources based on user demand.

Cloud Architecture

- Deploy application servers across multiple cloud regions.
- Store video files in cloud object storage.
- Use a Content Delivery Network (CDN) to cache videos near users.
- Use managed databases for user information and streaming metadata.
- Implement cloud monitoring services for performance tracking**.**

Load Balancing and Auto-Scaling

- Load balancers distribute user requests evenly among multiple servers.
- Auto-scaling automatically adds servers during peak traffic.
- Removes unnecessary servers during low traffic.

- Prevents server overload.
- Ensures uninterrupted video streaming.

Cost Optimization

- Use pay-as-you-go pricing.
- Enable automatic scaling to avoid unused resources.
- Store rarely accessed videos in low-cost archival storage.
- Use CDN caching to reduce bandwidth costs.
- Continuously monitor cloud resource usage.

Conclusion

A cloud architecture with CDN, load balancing, auto-scaling, and monitoring provides high performance, minimizes buffering, improves user experience, and reduces operational costs.

Question 2

A fintech startup is deploying applications using containers across multiple cloud providers. They face challenges in maintaining consistency and managing deployments. Propose a solution using container orchestration and multi-cloud strategies. Explain how service discovery and scaling can be handled efficiently. Highlight the benefits and risks of a multi-cloud deployment model.

Answer

Introduction

Containers provide lightweight application deployment, while multi-cloud strategies improve reliability. Container orchestration ensures consistent deployment and management across different cloud platforms.

Proposed Solution

- Package applications using containers.
- Deploy containers using Kubernetes.
- Use Infrastructure as Code for consistent deployment.
- Maintain identical configurations across all cloud providers.

- Automate deployments using CI/CD pipelines.

Service Discovery and Scaling

- Kubernetes automatically discovers services.
- Internal DNS allows containers to communicate easily.
- Horizontal Pod Autoscaler increases or decreases container instances based on demand.
- Load balancers distribute traffic evenly.
- Self-healing automatically restarts failed containers.

Benefits

- High availability.
- Vendor independence.
- Better disaster recovery.
- Improved scalability.
- Flexible workload distribution.

Risks

- Higher management complexity.
- Increased security challenges.
- Higher networking costs.
- Data synchronization issues.
- Complex monitoring across providers.

Conclusion

Using Kubernetes with multi-cloud deployment provides scalable, consistent, and reliable application management while reducing vendor lock-in.

Question 3

An e-learning platform experiences data loss due to accidental deletion in the cloud. The organization needs a robust backup and disaster recovery strategy. Design a solution using cloud-native backup, replication, and versioning. Explain how Recovery Time Objective (RTO) and Recovery Point Objective (RPO) are achieved. Provide steps to ensure business continuity during failures.

Answer

Introduction

Accidental deletion can lead to serious data loss. A cloud-native backup and disaster recovery solution protects important educational content and ensures rapid recovery.

Backup Strategy

- Schedule automatic cloud backups.
- Store backups in multiple geographic regions.
- Enable object versioning.
- Replicate data continuously.
- Perform regular backup verification.

Achieving RTO and RPO

- Low RTO is achieved using standby recovery servers and automated restoration.
- Low RPO is achieved through continuous data replication and frequent backups.
- Backup snapshots reduce data loss.
- Automated failover speeds up recovery.
- Regular testing validates recovery procedures.

Business Continuity Steps

- Maintain disaster recovery plans.
- Monitor backup health.
- Test restoration procedures.
- Train administrators regularly.

- Maintain redundant cloud infrastructure**.**

Conclusion

Cloud backup, replication, and versioning protect against accidental deletion while ensuring fast recovery and continuous service availability.

Question 4

A government agency is adopting a hybrid cloud model for sensitive and non-sensitive data. Some applications must remain on-premises due to compliance requirements. Design a hybrid architecture that ensures secure communication between environments. Explain how identity management and encryption should be implemented. Discuss challenges in managing hybrid cloud environments.

Answer

Introduction

A hybrid cloud combines on-premises infrastructure with public cloud services. Sensitive government data remains on-premises, while less sensitive workloads are deployed in the public cloud**.**

Hybrid Cloud Architecture

- Store confidential data in the private data center.
- Deploy public services in the public cloud.
- Connect both environments using secure VPN or dedicated links.
- Use centralized cloud management tools.
- Implement network segmentation.

Identity Management

- Use Federated Identity Management.
- Implement Single Sign-On (SSO).
- Enable Multi-Factor Authentication (MFA).
- Apply Role-Based Access Control (RBAC).
- Monitor user activities continuously.

Encryption

- Encrypt stored data.
- Encrypt data during transmission using TLS.
- Secure encryption keys.
- Perform regular security audits.
- Apply strict access policies.

Challenges

- Complex infrastructure management.
- Compliance monitoring.
- Network latency.
- Security policy consistency.
- Higher operational complexity.

Conclusion

Hybrid cloud architecture provides flexibility while meeting government compliance requirements through secure communication, identity management, and encryption.

Question 5

A SaaS provider needs to ensure high availability and fault tolerance for its application. The system must continue operating even if one region goes down. Propose a multi-region deployment strategy in the cloud. Explain how data replication and failover mechanisms should be configured. Discuss monitoring and alerting strategies for proactive issue detection.

Answer

Introduction

High availability ensures applications remain accessible even during failures. A multi-region cloud deployment provides fault tolerance by distributing workloads across geographically separated regions.

Multi-Region Deployment

- Deploy application servers in multiple cloud regions.
- Use global load balancers.
- Synchronize databases across regions.
- Distribute user traffic intelligently.
- Maintain standby infrastructure.

Data Replication and Failover

- Continuously replicate databases between regions.
- Configure automatic failover.
- Maintain synchronized backups.
- Regularly test failover procedures.
- Ensure minimal service interruption.

Monitoring and Alerting

- Continuously monitor server health.
- Track CPU, memory, and network usage.
- Configure real-time alerts.
- Analyze logs for unusual activities.
- Automatically notify administrators of failures.

Conclusion

A multi-region deployment with data replication, automatic failover, and continuous monitoring provides high availability, fault tolerance, and uninterrupted SaaS application services.